Matrix Chain Multiplication

Ques- $A_1 \cdot A_2 \cdot A_3 \cdot A_4$

$5\times\underline{4}$ $\underline{4}\times\underline{6}$ $\underline{6}\times\underline{2}$ $\underline{2}\times 7$

✓ same ✓ same ✓ same

Step 1 Draw two Matrix of how much Matrix are given ($n=4 \rightarrow A_1, A_2, A_3, A_4$)

m.

	1	2	3	4
1	0	120	88	158
2		0	48	104
3			0	84
4				0

S

	1	2	3	4
1		1	1	3
2				3
3				3
4				

$m[1,1]$
$A_1 \cdot A_1$
$= 0$

$m[2,2]$
$A_2 \cdot A_2$
$= 0$

$m[1,2]$
$A_1 \cdot A_2$
5×4 4×6
Total multip seq $= 5\times4\times6$
$= 120$

$m[2,3]$
$A_2 \cdot A_3$
$4\times\underline{6}$ 6×2
Total mult seq $= 4\times6\times2$
$= 48$

$m[3,4]$
$A_3 \cdot A_4$
6×2 2×7
Total mult seq $= 6\times2\times7$
$= 84$

$m[1,3]$

$A_1 \cdot A_2 \cdot A_3$

$A_1 \cdot (A_2 \cdot A_3)$ $5\times\underline{4}$ 4×6 $6\times\underline{2}$

$(A_1 \cdot A_2) A_3$ $5\times\underline{4}$ $4\times\underline{6}$ $6\times\underline{2}$

minimum

$m[1,1] + m[2,3] + 5\times4\times2$
$0 + 48 + 40$
$= 88$

$m[1,2] + m[3,3] + 5\times6\times2$
$120 + 0 + 60$
$= 180$

88 is minimum so update in m & we have taken 1st approach $(A_1 \cdot A_2)A_3$ for where we get 88
So update 1 on S matrix

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$m[2,4]$

$A_2 \cdot (A_3 \cdot A_4)$

$\underline{4\times6}$ 6×2 2×7

$m[2,2] + m[3,4] + 4\times6\times7$

$0 + 84 + 168$

$= 252$

$(A_2 \cdot A_3)\, A_4$

$\underline{4\times6}$ 6×2 $\underline{2\times7}$

$m[2,3] + m[4,4] + 4\times2\times7$

$48 + 0 + 56$

$= 104$ ← minimum so update 104 at m

& Approach $(A_2 \cdot A_3)\, A_4$ → add 3 at S note

$m[1,4] = \text{minimum}\{ \overset{0}{m[1,1]} + \overset{140}{m[2,4]} + \overset{140}{5\times4\times7}$, $\underset{120}{m[1,2]} + \underset{84}{m[3,4]} + \underset{210}{5\times6\times7}$,

$A_1 \cdot (A_2 \cdot A_3 \cdot A_4)$

$\underset{88}{m[1,3]} + \underset{0}{m[4,4]} + \underset{70}{5\times2\times7} \}$

→ at this value it gives answer fill it in S

$88 + 0 + 70 = 158$ fill it in m

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